Monthly Intelligence Report

Edition 001 — The North West-West Midlands Seismic Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS	PUBLIC DATA SOURCES
3,150	~22,765	35.3 M	30+
342 EHV · 2,808 HV	~87,000 km coverage	10,000 per substation	95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 3,150 substations and ~22,765 grid lines spanning ~105,000 km.

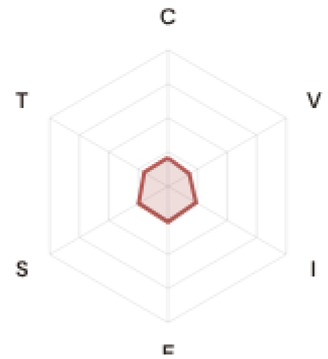
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering the UK's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation Outer London 0090

Outer London, Greater London · 66 kV

0.184	Low	0.063	3th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.206 / 0.30 (69%) ⚠

I Infrastructure: 0.235 / 0.25 (94%) ⚠

S Saturation: 0.247 / 0.20 (123%)

V Voltage: 0.180 / 0.10 (180%)

E Economic: 0.260 / 0.10 (260%)

T Transition: 0.200 / 0.05 (400%)

MODIFIERS

R3 Consequence: 0.937 ↓ dampens

R4 Graph Criticality: 0.863 ↓ dampens

R6a Restoration: 1.001 → neutral

R6b Seismic: 1.000 → neutral

R7 Digital Readiness: 1.022 ↑ amplifies

This month's spotlight — automatically selected for April 2026 — is **Substation Outer London 0090** in Outer London, Greater London. With an R_final of 0.184 (Low band, 3th fleet percentile), its risk profile is dominated by the T (Transition) component at 400% of its weight ceiling, followed by E (Economic) at 260%. Modifiers collectively shift R_base by 17.9%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that the UK's Digital Strategy / Net Zero Strategy planning requires.

CYBER HIGH EXPOSURE

1790

56.8% of fleet

BLIND SPOTS

291

High/Critical + R7 < median

AVG R7 MODIFIER

1.0041

Range [0.99, 1.05]

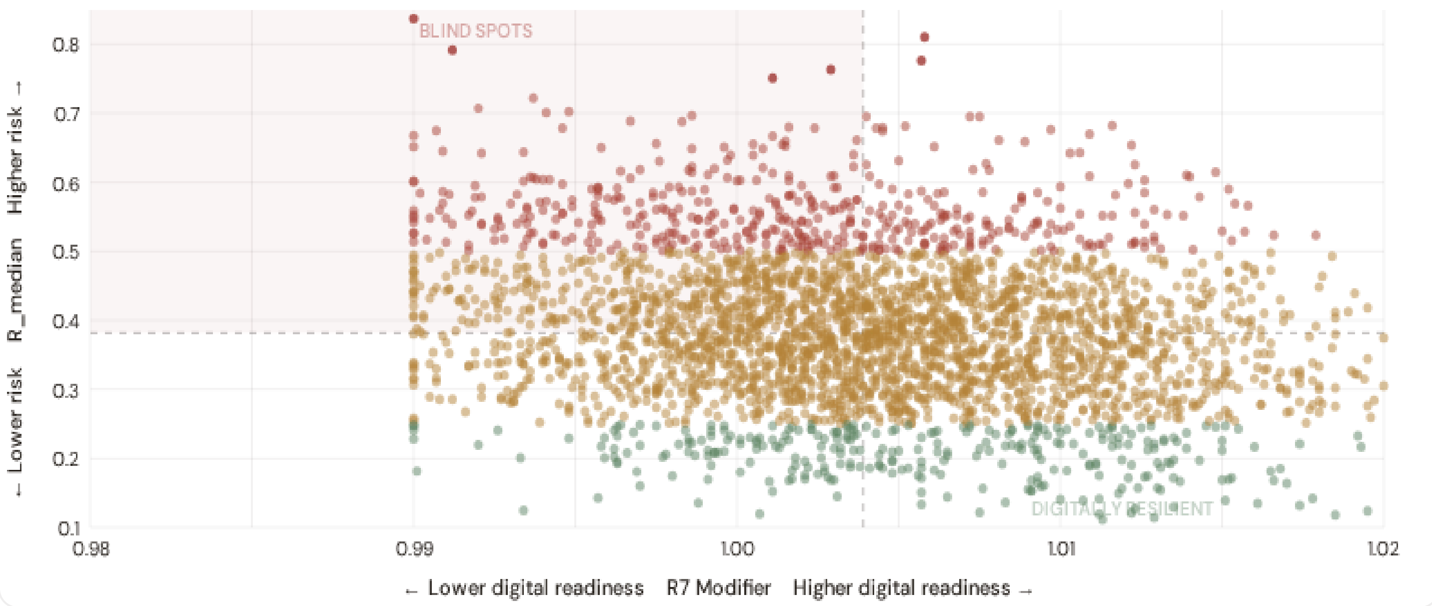
HIGH-CONSEQUENCE SUBS

66

2.1% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **291 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0039). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Wales (78), North East (39), North West (36)**. Across the full fleet, 1790 substations (56.8%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in major metropolitan areas (London, Manchester, Birmingham).

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

788 substations (25.0%) High/Critical: **190** (24.1%) Avg R: **0.421** Avg E: **0.427** / 0.10

Low 56 Med 542 High 184 Crit 6

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

789 substations (25.0%) High/Critical: **130** (16.5%) Avg R: **0.402** Avg E: **0.431** / 0.10

Low 48 Med 611 High 130 Crit 0

🏢 Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

786 substations (25.0%) High/Critical: **99** (12.6%) Avg R: **0.377** Avg E: **0.430** / 0.10

Low 84 Med 603 High 99 Crit 0

🌾 Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

787 substations (25.0%) High/Critical: **47** (6.0%) Avg R: **0.344** Avg E: **0.404** / 0.10

Low 118 Med 622 High 47 Crit 0

Value of Lost Load (VoLL) context: The ACER/CEPA 2022 pan-European VoLL study estimates the UK's sector-weighted average at approximately £8.0/kWh, with industrial customers at ~£14.2/kWh and residential at ~£4.5/kWh. The SSI's R3 consequence multiplier maps these sectoral weights to each substation's service territory. **17 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **North West (99), Yorkshire (81), South West (76)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of £10–20/kWh, compared to £1–3/kWh in rural agricultural zones. The fleet-wide average energy poverty rate is 14.5%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 County Digital Readiness Ranking

Countys ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify counties combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 COUNTIES

Longer bar = higher R7 = better digital infrastructure.

1	Dyfed ⚠ high R	0.9947
2	Powys ⚠ high R	0.9951
3	South Glamorgan ⚠ high R	0.9958
4	Gwent ⚠ high R	0.9959
5	Clwyd ⚠ high R	0.9961
6	Gwynedd ⚠ high R	0.9963
7	Antrim ⚠ high R	0.9967
8	Down ⚠ high R	0.9971
9	Fermanagh ⚠ high R	0.9976
10	Cleveland ⚠ high R	0.9984
11	Armagh ⚠ high R	0.9985
12	Northumberland ⚠ high R	0.9989
13	Londonderry ⚠ high R	0.9991
14	County Durham ⚠ high R	0.9992
15	Tyne and Wear ⚠ high R	0.9993

County-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Dyfed** has the lowest average R7 (0.9947), while **Outer London** leads at 1.0136 — a spread of 0.0189 across the R7 range. **33 counties** combine below-median digital readiness with above-median grid risk, making them priority targets for Net Zero 2050 / Levelling Up Fund digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0041, median = 1.0039; 1790 substations classified HIGH cyber-exposure; 291 blind-spot substations (High/Critical risk + below-median R7); 4 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging counties where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL

30

95 variables

4

High-frequency feeds

18

Regular refresh cycle

STATIC / DERIVED

8

Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
ONS-LM	ONS Labour Market Statistics	MONTHLY	County	1 var
DESNZ	BEIS/DESNZ (Dept for Energy Security and Net Zero)	QUARTERLY	County	5 vars
OFGEM	Ofgem (Office of Gas and Electricity Markets)	ANNUAL	County	8 vars
ONS	ONS (Office for National Statistics)	ANNUAL	County	8 vars
DEFRA	DEFRA/EA (Dept for Environment, Food & Rural Affairs)	ANNUAL	regions	4 vars
JRC	JRC DNO Observatory	ANNUAL	DNO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
NCSC	NCSC (National Cyber Security Centre)	ANNUAL	National	2 vars
EA-FR	EA Flood Risk / Cabinet Office CCA	ANNUAL	County	2 vars
ONR	ONR (Office for Nuclear Regulation)	ANNUAL	Site-level	1 var
ONS-E	ONS Economic Statistics	ANNUAL	regions	3 vars
NGESO-S	NGESO Statistical Series	ANNUAL	regions	2 vars
NR	Network Rail Statistics	ANNUAL	Countyl	1 var
DFT	DfT/DVLA (Driver and Vehicle Licensing Agency)	ANNUAL	Countyl	1 var
OFGEM-M	Ofgem RIIO Monitoring Report	ANNUAL	DNO-level	2 vars
BGS	BGS (British Geological Survey)	STATIC	County	3 vars
CDS	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
BGS-SH	BGS Seismic Hazard Assessment	STATIC	Municipal	1 var
ONS-GEO	ONS Geography Registry	STATIC	Countyl	1 var
ISO9223	ISO 9223 Corrosion	DERIVED	County	1 var

NGESO	NGESO (National Grid ESO) — Data Portal	HOURLY	Bidding zone	**3 vars**
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	**2 vars**
OM	Open-Meteo Historical Weather	HOURLY	~1 km	**3 vars**
METOFF	Met Office (UK Met Office)	HOURLY	Station	**3 vars**

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (2 sources) — Grid load + procurement data

BEIS/DESNZ Registry · NGESO Transparency

Quarterly (1 sources) — DER registry + business data + macro indicators

DFT

Annual (17 sources) — Core grid quality + demographics + environment

OFGEM · ONS · DEFRA · ONS-LM · DESNZ · NGESO-S · OFGEM-M · EEA · Eurostat · DESI · NCSC · ONR · JRC · Cabinet Office · NR · METOFF · ONS-E

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · BGS · ISO 9223 · ONS/OS · BGS Seismic Hazard

C.2 Fleet Score Snapshot

FLEET SIZE
3,150
substations scored

MEDIAN R
0.382
P5=0.214 · P95=0.571

HIGH + CRITICAL
466
14.8% of fleet

HIGH CONFIDENCE
100%
3150 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 306 (9.7%) · Medium 2378 (75.5%) · High 460 (14.6%) · Critical 6 (0.2%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **3,150 substation scores remain unchanged** this edition. The fleet median R of 0.382 (mean 0.386) reflects a distribution skewed toward the lower bands, with 9.7% in Low and 75.5% in Medium. The first material score changes are expected when Ofgem publishes the annual TIQE indicators (affecting C1–C4 and R6a) and when BEIS/DESNZ refreshes the Renewable Energy Planning Database (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — County-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — R&D + economic structure	§6
L3	ENHANCED	V_socio Fiscal Enrichment — ONS + DEFRA/EA + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — ONS net migration	§5
L5	DATA	95/95 variables operational (100%). 30 data sources total.	§8, §12
G1	DATA	BEIS/DESNZ upgraded to quarterly registry — County-level DER registry	§12
G2	DATA	BGS upgraded to live API — County-level seismic data	§12
G3	DATA	OSM upgraded to Overpass API — 3,150 real substations	§12
G4	NEW	R6b Seismic Hazard modifier — BGS Seismic Hazard classification from BGS	§5
G5	NEW	Network & Topology layer (H): 7 variables — NGESO Grid Plan, seismic analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The North West-West Midlands Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** North West-West Midlands seismic corridor · **002** DER saturation hotspots (East Anglia/West Midlands) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Nuclear fleet dependency and phase-out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the North West-West Midlands, Why Now

NORTH WEST-WEST MIDLANDS SUBSTATIONS

340

151 EHV · 189 HV

HIGH BAND

53

15.6% of corridor

CORRIDOR MEDIAN R

0.403

Fleet median: 0.382

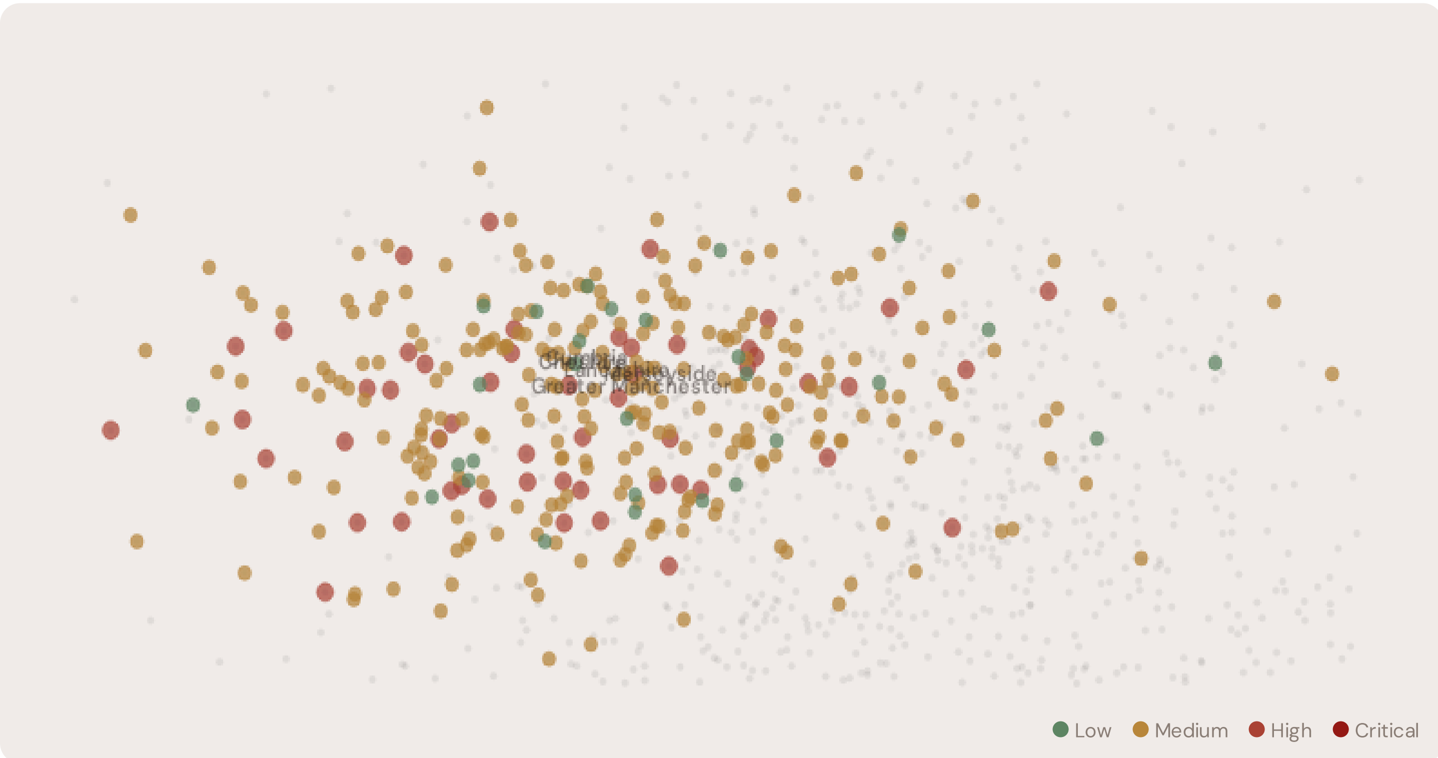
WORST COUNTY

Lancashire

Avg R: 0.416 · 68 substations

The North West-West Midlands hosts **340 substations** across six counties, making it one of the most energy-critical corridors in Europe. However, its risk profile is disproportionately concentrated in the upper bands: **53 substations (15.6%)** are classified High, with only **28** in the Low band. 56% of North West-West Midlands substations score above the national fleet median of 0.382. The corridor median R of 0.403 reflects the tension between seasonal demand volatility driven by tourism and agriculture, rapidly growing solar and wind capacity, and aging grid infrastructure not dimensioned for bidirectional renewable flows. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under UK 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	COUNTY	R_FINAL	BAND	C	V	I	E	S	T	ALERT
GSP Cheshire 2289	Cheshire	0.695	High	0.551	0.686	0.853	0.562	0.730	0.681	C, I

SUBSTATION	COUNTY	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Grid Supply Point Lancashire 2396	Lancashire	0.695	High	0.402	0.415	0.842	0.857	0.394	0.782	C, I
Substation Cumbria 2235	Cumbria	0.688	High	0.787	0.410	0.753	0.556	0.619	0.550	C, I
Grid Supply Point Lancashire 2161	Lancashire	0.679	High	0.575	0.589	0.436	0.553	0.577	0.459	C, I, R4
Primary Merseyside 2098	Merseyside	0.661	High	0.480	0.425	0.599	0.579	0.524	0.461	C, I
BSP Greater Manchester 2202	Greater Manchester	0.655	High	0.588	0.428	0.757	0.785	0.423	0.547	C, I
GSP Cheshire 2359	Cheshire	0.653	High	0.491	0.865	0.479	0.545	0.904	0.469	C, I
BSP Greater Manchester 2362	Greater Manchester	0.627	High	0.596	0.328	0.576	0.576	0.414	0.788	C, I
Substation Cumbria 2255	Cumbria	0.623	High	0.694	0.682	0.578	0.391	0.859	0.348	C, I
Substation Cumbria 2250	Cumbria	0.623	High	0.705	0.642	0.577	0.273	0.433	0.709	C, I

… 330 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — NORTH WEST-WEST MIDLANDS VS FLEET

C — Continuity	0.436 vs 0.418 (+4%)
I — Infrastructure	0.430 vs 0.418 (+3%)
S — Saturation	0.427 vs 0.424 (+1%)
V — Voltage	0.430 vs 0.427 (+1%)
E — Economic	0.441 vs 0.423 (+4%)

MODIFIER AVERAGES — NORTH WEST-WEST MIDLANDS VS FLEET

R3 Consequence	1.008	fleet 0.998	→
R4 Graph Crit.	0.915	fleet 0.916	↓
R6a Restoration	1.000	fleet 0.999	→
R6b Seismic	1.001	fleet 1.001	→
R7 Digital Read.	1.002	fleet 1.004	→

T — Transition 0.438 vs 0.426 (+3%)

The dominant risk driver across the North West-West Midlands corridor is **T (Transition)**, averaging 0.438 against a fleet average of 0.426 — occupying 876% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.441 vs fleet 0.423 (441% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.008 (fleet: 0.998), reflecting the socio-economic fragility of the North West-West Midlands corridor's agriculture-intensive, tourism-dependent economy with growing renewable capacity but limited grid interconnection. The R6b seismic modifier averages 1.001, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 County Breakdown

COUNTY RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Lancashire 68 subs · avg R = 0.416 · H:15 M:49

Cumbria 68 subs · avg R = 0.407 · H:8 M:56

Greater Manchester 68 subs · avg R = 0.406 · H:12 M:52

Cheshire 68 subs · avg R = 0.390 · H:11 M:50

Merseyside 68 subs · avg R = 0.378 · H:7 M:52

The county-level breakdown reveals significant intra-regional variation. **Lancashire** leads with a mean R of 0.416 across 68 substations, while **Merseyside** scores 0.378 — a spread of 0.038 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the North West-West Midlands is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

7 North West-West Midlands substations score R ≥ 0.650, placing them in the upper quartile of national risk. For Net Zero Strategy investment prioritisation, this analysis identifies the Lancashire corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

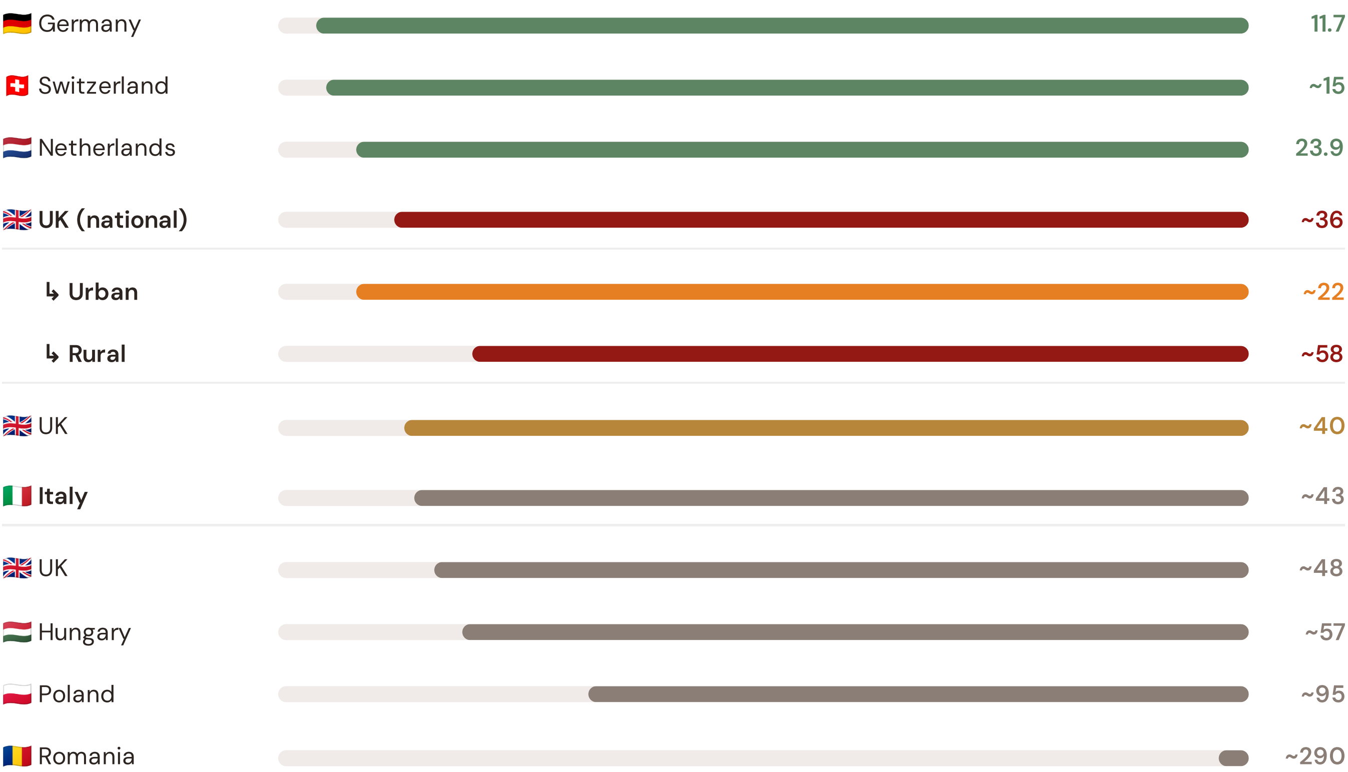
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the North West-West Midlands corridor analysis hinges on the UK's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected EU Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



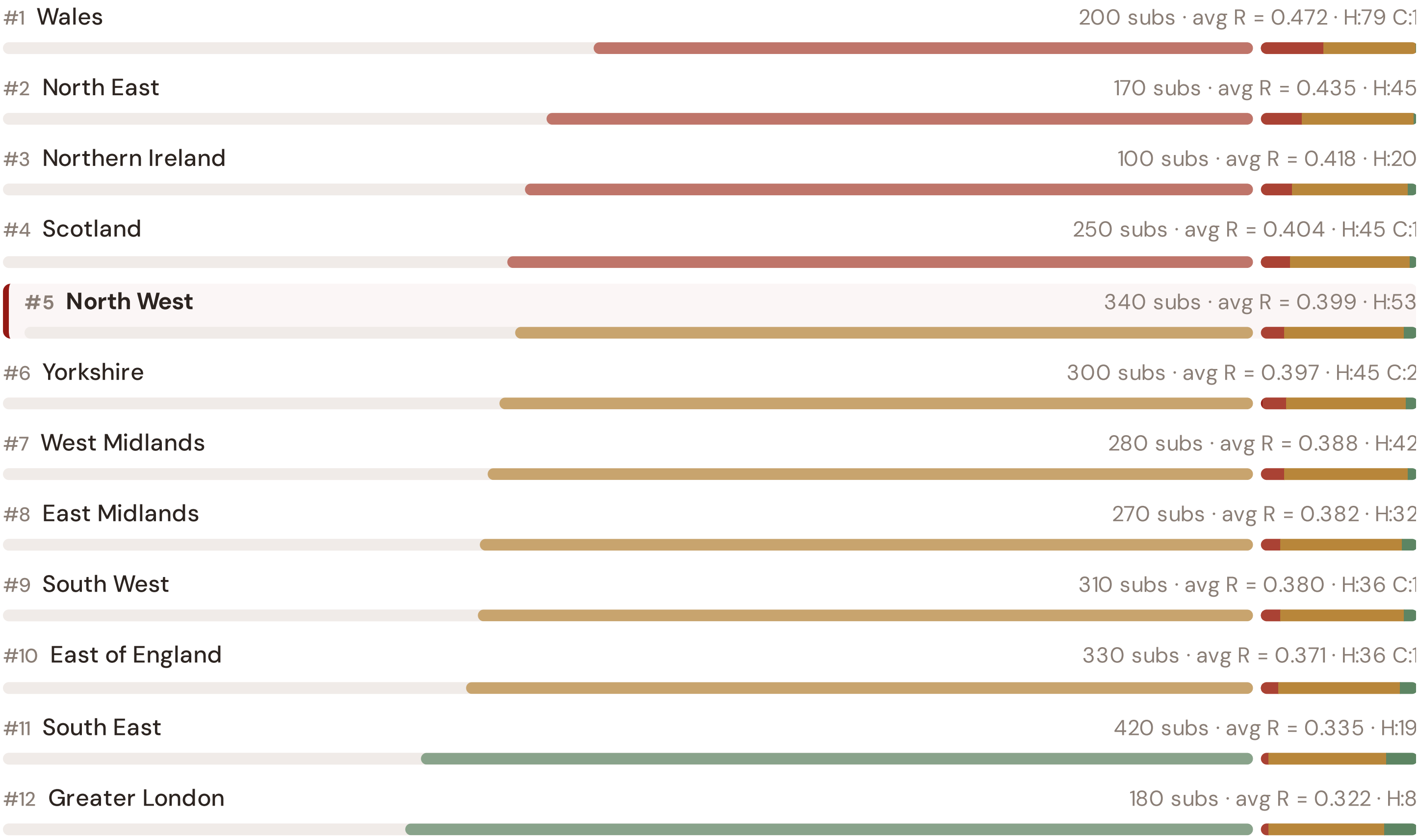
Source: CEER 7th Benchmarking Report (2022); Ofgem TIQE (2023); Bundesnetzagentur (2024). UK sub-categories from Ofgem territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The North West-West Midlands corridor deep-dive shows **326 substations** (of 340) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with the UK's rural territory classification.The North West-West Midlands's average C of 0.436 is **1.0× the fleet average** of 0.418. In European terms, these substations individually perform closer to Hungarian or European SAIDI levels (~48–57 min/customer/year) than to the UK's own national average (~36 min).The SSI reveals what aggregate CEER statistics conceal: that the UK's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the North West-West Midlands corridor concentrates the worst of the latter.

Regional Risk Ranking — All 17 Regions

Where does the North West-West Midlands sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The North West-West Midlands ranks **#0 of 12 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Wales, North East, Northern Ireland, while the three most resilient are Greater London, South East, East of England. 7 of 12 regions score above the fleet average of 0.386. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DNO or national statistics — reveals the true spatial distribution of grid stress across UK. It is precisely this substation-level granularity that positions the SSI as a tool for Net Zero Strategy / Levelling Up Fund investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the NGESO/DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy,	NGESO/DNO + Community value		Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

normalisation, MC engine	stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP		

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a £10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against Ofgem (Office of Gas and Electricity Markets)'s published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The British training set comprises 3,150 substations with complete 95-feature vectors. Average CI_width of 0.134 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 306 Low, 2378 Medium, 460 High, 6 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The Wales Solar Frontier

Deep-dive into Wales's grid risk profile — substation map, component analysis, county breakdown. European Context Card: Massive PV build-out vs weak transmission capacity and rural grid topology.

NEXT DATA REFRESH

Q3 2026 — 1 Quarterly Sources

DFT are due for refresh in Q3. Impact: BEIS/DESNZ (Renewable Energy Planning Database) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Ofgem (Office of Gas and Electricity Markets)** (8 variables → C1–C4 (SAIDI/SAIFI/CAIDI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Wales**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a British utility, Ofgem, NGESO, regional DNO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 3,150 substations · ~22,765 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5